

数学作业纸

科目_____

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$$7.2.7 (1) I_C = I_{C1} = I_{C2} = \frac{I_0}{2} = 0.5 \text{ mA} \quad \nabla$$

$$V_E = V_B - V_{BE} = -0.7 \text{ V}$$

$$V_{CE1} = V_{CE2}, V_{CE} = I_C R_{C1} - V_E = 5.7 \text{ V}$$

$$(2) r_{be1} = 200 + (1+\beta) \frac{26 \text{ mV}}{I_C} = 1057 \Omega$$

$$A_{vd} = -\frac{\beta R_{C1}}{r_{be1}} = -186.9$$

$$\text{单端输出时, } A_{vd1} = \frac{A_{vd}}{2} = -93.45$$

$$A_{vc1} = -\frac{\beta R_{C1}}{r_{be1} + (1+\beta) R_E} \approx -0.2$$

$$K_{CMR1} = \left| \frac{A_{vd1}}{A_{vc1}} \right| = 467.25$$

$$7.2.8 (1) I_{E1} = I_{C1} = \frac{1}{2} I_0 = 1 \text{ mA}$$

$$r_{be1} = 200 + (1+\beta) \frac{26 \text{ mV}}{I_{E1}} = 2.81 \text{ k}\Omega$$

$$A_{vd} = -\frac{\beta R_{C1}}{r_{be1} + (1+\beta) R_E} = -43.41, V_O = V_i A_{vd} = -43.41 \times 0.02 = -0.87$$

$$(2) A_{vd} = -\frac{\beta (R_{C1} // \frac{R_L}{2})}{r_{be1} + (1+\beta) R_E} = -14.5$$

$$V_O = -0.29 \text{ V}$$

$$13.7 A_{vd2} = -\frac{1}{2} A_{vd} = 21.7$$

$$A_{vc2} = -\frac{\beta R_L}{r_{be1} + (1+\beta) (R_{E1} + 2R_O)} = -0.028$$

$$K_{CMR2} = \left| \frac{A_{vd2}}{A_{vc2}} \right| = 775$$

$$(4) R_{id} = 2[r_{be1} + (1+\beta) R_{E1}] = 25.8 \text{ k}\Omega$$

$$R_{ic} = \frac{1}{2} [r_{be1} + (1+\beta) (R_{E1} + 2R_O)] = 10.1 \text{ M}\Omega$$

$$R_{O2} = R_E = 5.6 \text{ k}\Omega$$

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7.12.9, (1) $V_E = -0.17V$.

$$I_{C1} = I_{C2} = \frac{1}{2} \frac{V_E + V_{BE}}{R_E} = 0.52mA.$$

$$I_{B1} = I_{B2} = \frac{I_C}{\beta} = 0.1mA.$$

$$V_{CE1} = V_{CE2} = V_{CC} - R_{C1} I_{C1} - V_E = 4V.$$

(2) $V_o = I_M R_M = 0.12V$.

$$r_{be} = r_{bb'} + (1+\beta) \frac{26V}{I_{C1}} = 2.85k\Omega.$$

$$A_{vd} = - \frac{\beta(R_{C1} // R_E)}{r_{be} + R_{B1}} = -3.25$$

$$V_{id} = \frac{V_o}{A_{vd}} = -0.037V.$$

$$V_{d1} = -0.031V, V_{d2} = 0.031V$$